

No.	Tumor	Organization / Guideline	Notes
1	Hepatocellular carcinoma (HCC)	AASLD: Practice Guidance — Hepatocellular Carcinoma	Prevention, screening, diagnosis, local/surgical/systemic therapy, follow-up
2	Hepatocellular carcinoma (HCC)	ACR LI-RADS	Imaging classification and reporting standard for high-risk populations (unified lexicon)
3	Hepatocellular carcinoma (HCC)	BCLC: Strategy for Prognosis & Treatment Allocation	BCLC staging and treatment allocation framework (widely cited)
4	Hepatocellular carcinoma (HCC)	EASL: Clinical Practice Guideline — Management of Hepatocellular Carcinoma	Screening, imaging/histologic diagnosis, local and systemic therapy
5	Hepatocellular carcinoma (HCC)	ESMO: Clinical Practice Guideline — Hepatocellular Carcinoma (2025)	Staging, conversion therapy, IO combinations and targeted strategies
6	Hepatocellular carcinoma (HCC)	NCCN: Hepatobiliary Cancers (2025)	Comprehensive management of HCC/ICC/biliary tract cancers
7	Lung cancer (NSCLC)	ACS: Lung Cancer Screening Guideline	Age 50–80; ≥ 20 pack-years; no ≥ 15 -year quit rule
8	Lung cancer (NSCLC)	USPSTF: Lung Cancer Screening Recommendation	Age 50–80; ≥ 20 pack-years; annual LDCT if quit < 15 years
9	Lung cancer (NSCLC)	ASCO: Perioperative IO/TKI Recommendations (2024)	Perioperative IO/TKI for early NSCLC
10	Lung cancer (NSCLC)	ASCO: Living Guideline — Stage IV NSCLC	Systemic therapy for stage IV (driver-positive and -negative)
11	Lung cancer (NSCLC)	ASCO: Rapid Recommendation — Stage III NSCLC	Stage III multimodal therapy (cCRT + consolidation IO, etc.)

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12	Lung cancer (NSCLC)	ESMO: Clinical Practice Guideline — Early & Locally Advanced NSCLC	Resectable/locally advanced; neoadjuvant/adjuvant IO/TKI
13	Lung cancer (NSCLC)	ESMO: Living Guideline — Oncogene-Addicted mNSCLC	Oncogene-addicted metastatic NSCLC
14	Lung cancer (NSCLC)	IASLC: Early NSCLC Multidisciplinary Consensus (2024/2025)	Staging imaging, surgical extent, perioperative systemic therapy
15	Lung cancer (NSCLC)	NCCN: Guidelines — Non–Small Cell Lung Cancer	From early to metastatic (molecular testing, perioperative IO/TKI, systemic therapy)
16	Lung cancer (SCLC)	NCCN: Guidelines — Small Cell Lung Cancer	Limited-stage cCRT; extensive-stage chemo + IO maintenance
17	Breast cancer	ASCO/CAP: HER2 Testing Guideline	Pathology standards (IHC/ISH)
18	Breast cancer	ASCO: Perioperative/Adjuvant Recommendations (2024)	Neoadjuvant/adjuvant chemo, endocrine and HER2-targeted therapy
19	Breast cancer	ASCO: Advanced HER2+ Breast Cancer — Systemic Therapy Guideline	Systemic therapy for metastatic HER2+ disease
20	Breast cancer	ESMO: Clinical Practice Guidelines — Early/Advanced Breast Cancer (2024/2025)	Full-course management for early and metastatic disease (clear evidence grading)
21	Breast cancer	ESMO: Clinical Practice Guideline — Early Breast Cancer	Early breast cancer (clear evidence grading)
22	Breast cancer	ESMO: Living Guideline — Metastatic Breast Cancer	Metastatic breast cancer (Living)

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23	Breast cancer	NCCN: Breast Cancer (2024/2025)	Subtype-driven therapy, surgery/radiation and follow-up
24	Hodgkin lymphoma (HL)	NCCN: Hodgkin Lymphoma	Diagnosis, staging (Cotswolds-modified Ann Arbor), 1st-line to relapse/refractory therapy and follow-up
25	B-cell lymphomas (DLBCL/MCL/FL/MZL, etc.)	NCCN: B-Cell Lymphomas	Comprehensive management and pathways for DLBCL, MCL, FL, MZL
26	T-cell lymphomas	NCCN: T-Cell Lymphomas	Classification and treatment pathways for PTCL/ALCL/ENKTL
27	Primary cutaneous lymphomas	NCCN: Primary Cutaneous Lymphomas	Key management points for cutaneous T/B-cell lymphomas (patient & professional versions)
28	Primary CNS lymphoma (PCNSL)	EHA–ESMO: Clinical Practice Guideline — Primary CNS Lymphomas	Diagnosis, imaging, systemic therapy, consolidation/maintenance
29	Diffuse large B-cell lymphoma (DLBCL) & others	ESMO: Haematological Malignancies — Lymphomas	Topic CPGs/algorithms for DLBCL, MCL, WM, primary cutaneous lymphomas, PTCL
30	Primary cutaneous lymphomas (CTCL/CBCL)	ESMO: Clinical Practice Guideline — Primary Cutaneous Lymphomas	Staging and treatment algorithms for MF/SS and other primary cutaneous lymphomas
31	DLBCL/LBCL	BSH: The Management of Newly Diagnosed Large B-Cell Lymphoma	Work-up, stratification and first-line regimens (incl. RT/autograft)
32	Marginal zone lymphoma (MZL)	BSH: Diagnosis and Management of Marginal Zone Lymphomas	Recommendations for extranodal MALT, splenic/nodal MZL

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33	Mantle cell lymphoma (MCL)	BSH: Diagnosis and Management of Mantle Cell Lymphoma	MCL stratification, transplant and novel agents
34	Follicular lymphoma (FL)	BSH: Guideline on the Investigation and Management of Follicular Lymphoma	Diagnosis, risk-adapted therapy and follow-up
35	Lymphoma (overview/follow-up)	JSH: Practical Guidelines for Hematological Malignancies — Lymphoma (Overview/Follow-up)	Post-treatment follow-up frequency and imaging
36	Non-Hodgkin lymphoma (adult)	NICE NG52: Non-Hodgkin's Lymphoma — Diagnosis and Management	Adult NHL diagnosis and management; related technology appraisals
37	Suspected cancer referral (incl. HL/NHL)	NICE NG12: Suspected Cancer — Recognition and Referral	Referral thresholds and timelines in primary care for suspected HL/NHL
38	Hodgkin lymphoma (HL)	DGHO/S3: Hodgkin Lymphoma (German)	German S3 evidence-based guideline: diagnosis, staging and treatment algorithms
39	Diffuse large B-cell lymphoma (DLBCL)	DGHO/Onkopedia: DLBCL (German)	German Onkopedia guideline: DLBCL management
40	Cutaneous T-cell lymphomas (MF/SS)	EORTC/ISCL/USCLC: 2023 Consensus Recommendations (MF/SS)	Treatment roadmap integrating systemic and skin-directed therapies

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3 *Abbreviations:* AASLD, American Association for the Study of Liver Diseases; ACR, American College of Radiology;
4 BCLC, Barcelona Clinic Liver Cancer; EASL, European Association for the Study of the Liver; ESMO, European Society
5 for Medical Oncology; NCCN, National Comprehensive Cancer Network; ACS, American Cancer Society; USPSTF, U.S.
6 Preventive Services Task Force; IASLC, International Association for the Study of Lung Cancer; CAP, College of American
7 Pathologists; EHA, European Hematology Association; BSH, British Society for Haematology; JSH, Japanese Society of
8 Hematology; NICE, National Institute for Health and Care Excellence; DGHO, German Society for Haematology and
9 Medical Oncology; EORTC/ISCL/USCLC, European/International/US cutaneous lymphoma groups; IO, immuno-

10 oncology; cCRT, concurrent chemoradiotherapy; 1L, first-line; R/R, relapsed/refractory; MF/SS, mycosis fungoides/Sézary
11 syndrome; WM, Waldenström macroglobulinemia; MALT, mucosa-associated lymphoid tissue.

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1	Hepatocellular carcinoma (HCC)	Chinese Association of Traditional Chinese Medicine: Guideline for TCM Diagnosis and Treatment of Primary Liver Cancer	TCM pattern differentiation and integrative therapy recommendations
2	Hepatocellular carcinoma (HCC)	China Anti-Cancer Association (CACA): Clinical Practice Guideline for Hepatocellular Carcinoma	Full-course management integrating domestic evidence and practice
3	Hepatocellular carcinoma (HCC)	Chinese Expert Consensus on the Use of Immune Checkpoint Inhibitors in Liver Transplant Recipients	ICI use and rejection-risk management before/after transplant
4	Hepatocellular carcinoma (HCC)	Chinese Guideline for the Diagnosis and Treatment of HCC with Bile Duct Tumor Thrombus (BDTT)	BDTT subtyping and diagnostic/treatment pathway
5	Hepatocellular carcinoma (HCC)	Chinese Expert Consensus on Conversion and Perioperative Treatment for Primary Liver Cancer	Conversion/neoadjuvant and perioperative strategies
6	Hepatocellular carcinoma (HCC)	National Health Commission of China: Guideline for the Diagnosis and Treatment of Primary Liver Cancer	Screening, diagnosis, staging, surgery/ablation/intervention, systemic therapy, follow-up
7	Lung cancer (NSCLC)	Chinese Society of Clinical Oncology (CSCO): Guideline for the Diagnosis and Treatment of Non-Small Cell Lung Cancer	Staging, molecular testing, perioperative IO/targeted therapy, metastatic management
8	Lung cancer (screening)	National Health Commission of China: Lung Cancer Screening and Early Detection/Treatment Program	Annual LDCT for high-risk groups; graded management after positive findings

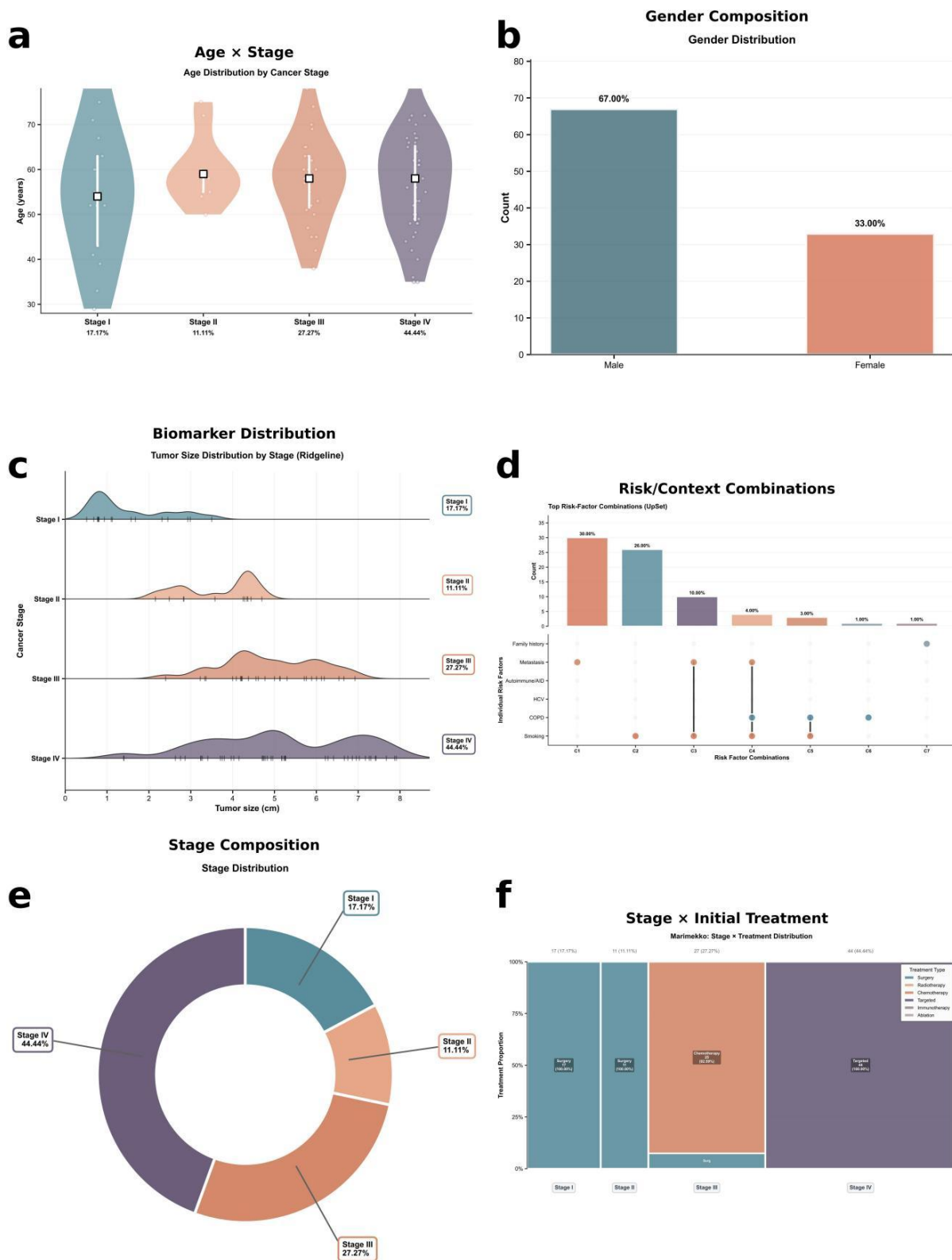
No.	Tumor	Organization / Guideline	Notes
9	Breast cancer	Chinese Society of Clinical Oncology (CSCO): Guideline for the Diagnosis and Treatment of Breast Cancer	China-specific, full-course management
10	Breast cancer	Chinese Medical Association: Guideline for the Diagnosis and Treatment of Breast Cancer	Screening, diagnosis, surgery and systemic therapy, follow-up
11	Breast cancer	Chinese Breast Cancer Society (CBCS): Essential Edition of the Breast Cancer Guideline	Key points for early and metastatic disease
12	Breast cancer	China Anti-Cancer Association: Guideline and Standards for the Diagnosis and Treatment of Breast Cancer	China-specific diagnostic and therapeutic standards
13	Lymphoma (comprehensive)	National Health Commission of China: Guideline for the Diagnosis and Treatment of Lymphoma	Adults/children HL & NHL; diagnosis, staging, treatment & follow-up (incl. radiotherapy, transplant)
14	Lymphoma (comprehensive)	Chinese Society of Clinical Oncology (CSCO): Guideline for the Diagnosis and Treatment of Lymphoma	Layered care / full-course management; covers HL, DLBCL, FL, MZL, MCL, PTCL, WM, PCNSL
15	Follicular lymphoma (FL)	Chinese Guideline for the Diagnosis and Treatment of Follicular Lymphoma	Diagnosis/staging, imaging/pathology points, initial/relapsed therapy and follow-up
16	Mantle cell lymphoma (MCL)	Chinese Guideline for the Diagnosis and Treatment of Mantle Cell Lymphoma	Stratified management for de novo and relapsed/refractory MCL; transplant/maintenance
17	Adult T-LBL/ALL	Chinese Expert Consensus on Diagnosis and Treatment of Adult T-LBL/ALL	Diagnostic pathway and treatment advice (incl. transplant)

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18	Peripheral T-cell lymphomas (PTCL)	Chinese Expert Consensus on Diagnosis and Treatment of PTCL	PTCL subtypes, staging and systemic therapy (incl. novel agents)
19	Extranodal NK/T-cell lymphoma, nasal type	Chinese Guideline/Consensus for Diagnosis and Treatment of NK/T-cell Lymphoma (Nasal Type)	L-asparaginase-based therapy, combined radiotherapy, and strategies for relapsed/refractory disease
20	Primary CNS lymphoma (PCNSL)	Chinese Expert Consensus/Guideline for the Diagnosis and Treatment of PCNSL	High-dose MTX–based regimens, radiotherapy/consolidation, and transplant strategy
21	Lymphoma imaging (PET/CT)	Chinese Society of Nuclear Medicine: Guideline on the Use of FDG-PET/CT and PET/MR in Lymphoma	Nuclear-medicine standards for staging/response/follow-up

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37 *Abbreviations:* CSCO, Chinese Society of Clinical Oncology; NHC, National Health Commission; CBCS, Chinese Breast
38 Cancer Society; ICI, immune checkpoint inhibitor; R/R, relapsed/refractory; L-asparaginase, L-asparaginase; PET,
39 positron emission tomography; MR, magnetic resonance.

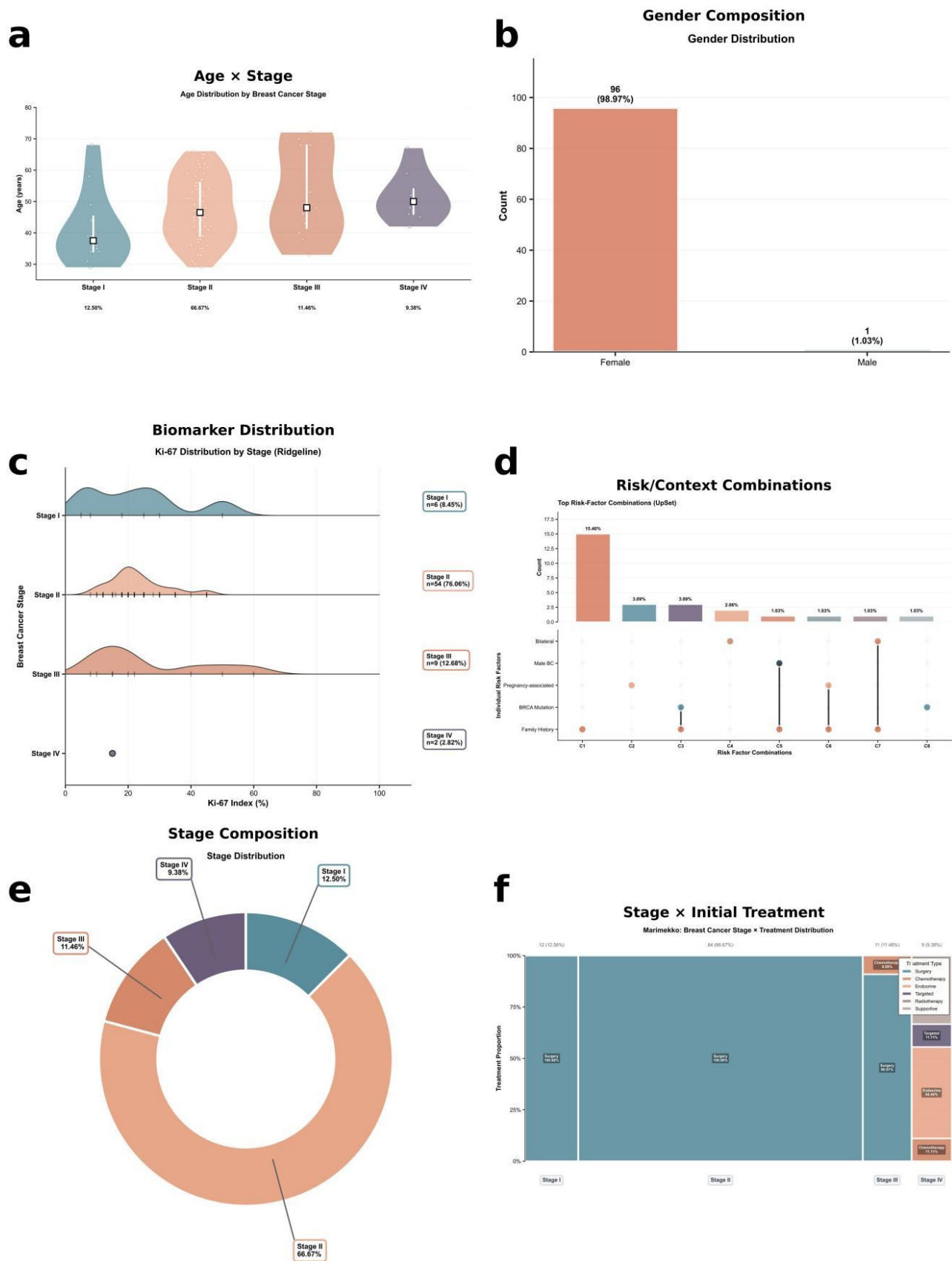
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42 Supplementary Figure 1. Lung Adenocarcinoma: Lesion-Level Design Cohort. Panel (a) depicts age distributions across
 43 TNM-8 clinical stages using violin plots, where white squares denote means, central bars indicate medians, and whiskers
 44 approximate 1.5×IQR. Stage labels display the percentage of the cohort mapped to each stage. Panel (b) shows the gender
 45 composition in bar format, illustrating male and female case counts with percentage labels annotated directly on the bars.
 46 Panel (c) visualizes lesion longest diameters (cm) stratified by stage through ridgeline plots, with tick marks along the x-

47 axis reflecting sample density and inter-stage variation. Panel (d) summarizes co-occurring clinical risk and context
48 combinations using an UpSet plot, highlighting intersections among key factors such as smoking history, COPD, and family
49 cancer predisposition. Panel (e) presents the overall stage composition in a donut chart, showing the proportional
50 representation of each TNM-8 stage with callout annotations. Finally, panel (f) displays the cross-distribution between stage
51 and initial treatment modality using a Marimekko chart, illustrating how early-stage patients primarily underwent surgical
52 resection ($\pm$ adjuvant therapy), while locally advanced and metastatic cases received multimodality or systemic regimens.
53 *Abbreviations:* COPD, chronic obstructive pulmonary disease; IQR, interquartile range; TNM-8, 8th edition TNM staging.

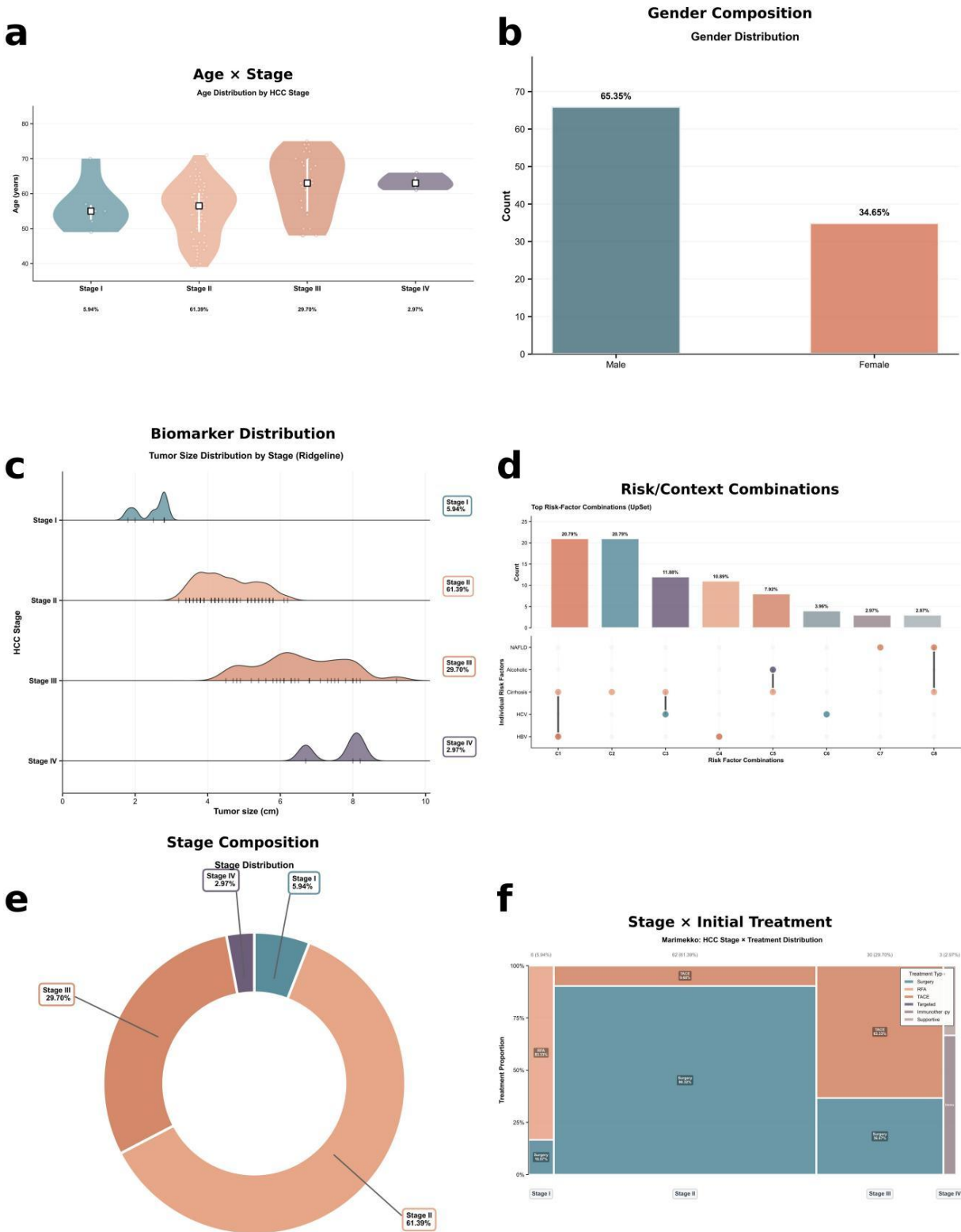


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55 Supplementary Figure 2. Breast Cancer: Lesion-Level Design Cohort. Panel (a) shows age distributions across clinical
 56 stages using violin plots, where white squares indicate the mean, central bars represent the median, and the overall spread
 57 reflects cohort heterogeneity. Panel (b) displays the gender composition in bar format, confirming the expected female

predominance with on-bar percentage labels. Panel (c) depicts stage-stratified biomarker distributions through ridgeline plots, highlighting representative quantitative indicators such as Ki-67 proliferation index or tumor size, with density tick marks reflecting intra-stage variability. Panel (d) summarizes risk and context combinations using an UpSet plot, emphasizing high-frequency intersections such as hereditary predisposition, pregnancy-associated breast cancer, and prior thoracic radiotherapy. Panel (e) presents overall stage composition in a donut chart, demonstrating the proportional share of each TNM or AJCC stage group at the time of index diagnosis. Finally, panel (f) visualizes the relationship between stage and initial treatment strategy using a Marimekko chart, outlining allocation across (neo)adjuvant pathways and systemic strategies for metastatic disease, stratified by biologic subtype logic (ER, PR, and HER2 status).

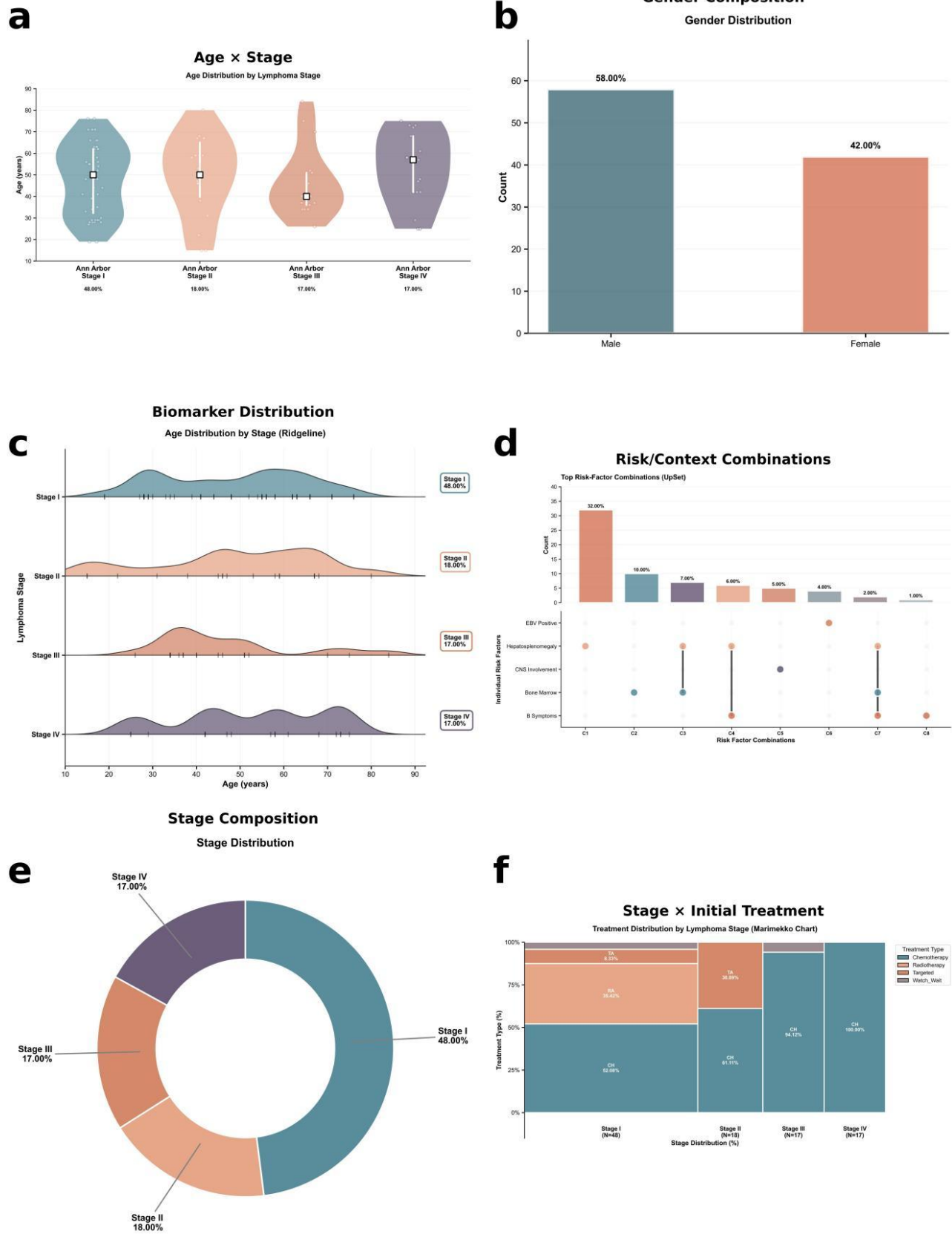
Abbreviations: ER, estrogen receptor; PR, progesterone receptor; HER2, human epidermal growth factor receptor 2; RT, radiotherapy.



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70 Supplementary Figure 3. Hepatocellular Carcinoma (HCC): Lesion-Level Design Cohort. Panel (a) shows age distributions
 71 across BCLC stages using violin plots, with white squares indicating means and bars denoting medians. Panel (b) presents
 72 gender composition with on-bar percentage labels, reflecting the expected male predominance. Panel (c) illustrates imaging-
 73 based lesion metrics by stage through ridgeline plots, using maximal tumor diameter (cm) as a representative parameter.
 74 Panel (d) depicts major etiologic and clinical contexts via an UpSet plot, highlighting intersections among HBV, HCV,

75 cirrhosis, portal hypertension, and NASH. Panel (e) shows stage composition in a donut chart, representing the proportion
76 of each BCLC stage at the index lesion. Panel (f) visualizes stage-specific initial treatment allocation in a Marimekko chart,
77 including resection, ablation, TACE/TARE, transplantation or bridging, and systemic therapy.
78 *Abbreviations:* BCLC, Barcelona Clinic Liver Cancer; HBV/HCV, hepatitis B/C virus; NASH, non-alcoholic steatohepatitis;
79 TACE, transarterial chemoembolization; TARE, transarterial radioembolization.



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82 Supplementary Figure 4. Lymphoma: Lesion-Level Design Cohort. Panel (a) shows age distributions by Ann Arbor stage
 83 using violin plots, where white squares indicate means and bars denote medians. Panel (b) displays gender composition
 84 with on-bar percentage labels. Panel (c) illustrates lesion burden metrics by stage through ridgeline plots, using

representative indicators such as the sum of product of diameters or the largest node size, with density ticks showing sample dispersion. Panel (d) summarizes co-occurring clinical contexts in an UpSet plot, highlighting intersections among B symptoms, bone marrow involvement, extranodal sites, and viral associations. Panel (e) depicts Ann Arbor stage distribution in a donut chart, while panel (f) visualizes treatment allocation by stage and histology in a Marimekko chart, including ABVD for Hodgkin lymphoma, R-CHOP for diffuse large B-cell lymphoma, and HD-MTX for primary CNS lymphoma.

Abbreviations: HL, Hodgkin lymphoma; DLBCL, diffuse large B-cell lymphoma; PCNSL, primary CNS lymphoma; ABVD, doxorubicin-bleomycin-vinblastine-dacarbazine; R-CHOP, rituximab-cyclophosphamide-doxorubicin-vincristine-prednisone; HD-MTX, high-dose methotrexate.